

Business Case: AI-Assisted Support Triage

A cost/impact analysis of deploying the Customer Support Triage Agent *Prepared as a standalone business analysis, built on top of an existing working prototype ([code](#)).*

1. The Business Problem

Support teams at consumer fintech companies (BNPL, digital payments) receive a high volume of tickets that fall into a small number of repeating categories: payment inquiries, disputes, technical issues, and — critically — suspected fraud. Two costs stack up:

- **Time cost:** agents spend time triaging every ticket manually before they can act on it, even when the correct routing is obvious from the message alone.
- **Risk cost:** fraud and high-risk disputes are only as fast as the slowest link in the queue — a delayed escalation on a fraud signal is a real financial exposure, not just an inconvenience.

Question this analysis answers: does an AI triage layer in front of the human queue pay for itself, and where does the value actually come from?

2. Assumptions (explicit, for transparency)

All figures below are **illustrative planning assumptions** for a mid-sized BNPL support operation, not measured data. They are chosen to be conservative and easy to challenge/replace with real numbers.

Assumption	Value	Rationale
Support tickets per month	12,000	Mid-sized BNPL operation, illustrative
Avg. manual triage time per ticket	3 minutes	Reading, classifying, deciding routing — before actual resolution work
Fully-loaded agent cost	SAR 35/hour	Riyadh support-agent market rate, illustrative

% of tickets that are low-ambiguity (clear category, no fraud/dispute signal)	55%	Typical share of routine inquiries in payments support
% of tickets flagged fraud/high-risk	4%	Conservative estimate for a payments platform
Avg. cost of a delayed fraud escalation (per incident, blended)	SAR 800	Chargeback + investigation + potential regulatory exposure, illustrative
Current avg. time-to-escalation for fraud tickets (manual queue)	45 minutes	Time sitting in a shared queue before a human notices the signal

Where real operational data exists, every number in this section should be replaced before using this analysis to justify a real spend decision.

3. What the AI Layer Actually Changes

Based on the working prototype, the triage agent does three things a human triager currently does manually:

1. **Classifies** the ticket into a category with a confidence score (~1–2 seconds, vs. ~3 minutes manual).
2. **Pulls customer context automatically** (account status, flags) — a lookup a human agent currently does as a separate step.
3. **Escalates immediately** on fraud signals, flagged accounts, or low confidence — removing the ticket from the general queue the moment the signal is detected, rather than waiting for a human to reach it.

It does **not** replace agents on resolution — disputes, technical issues, and anything escalated still go to a human. This is a triage layer, not an automation of judgment calls.

4. Cost Impact — Time Saved on Routine Triage

Routine tickets/month	= 12,000 × 55%	= 6,600
Manual triage time saved	= 6,600 × 3 minutes	= 19,800 minutes = 330
Monthly cost saved	= 330 hours × SAR 35/hour	= SAR 11,550/month
Annualized	=	SAR 138,600/year

This is the conservative, easy-to-defend number: it only counts the **triage step**, not resolution time, and only on the clearly-routine 55% of volume — the segment where automated classification is least likely to be wrong.

5. Risk Impact — Faster Fraud Escalation

This is the harder number to quantify, but arguably the more important one.

Fraud-flagged tickets/month = 12,000 × 4% = 480
Current avg. time-to-escalation = 45 minutes (manual queue)
Agent time-to-escalation = ~2 seconds (immediate, rule-based)

Converting escalation-speed improvement into a dollar figure requires knowing what fraction of the 45-minute delay is ever costly. Rather than invent a false-precision number, this is best framed as a **sensitivity range**:

If faster escalation prevents this % of would-be-costly delays	Monthly value	Annual value
2% of 480 tickets (~10/month)	SAR 8,000	SAR 96,000
5% of 480 tickets (~24/month)	SAR 19,200	SAR 230,400
10% of 480 tickets (~48/month)	SAR 38,400	SAR 460,800

Recommendation: treat the 2% scenario as the underwriting case (conservative, defensible) and the higher scenarios as upside — this is standard practice when a cost driver is directionally certain but not yet measured.

6. Combined View

Scenario	Time savings	Risk-reduction value	Total annual value
Conservative (2% fraud scenario)	SAR 138,600	SAR 96,000	SAR 234,600
	SAR		

Mid (5% fraud scenario)	138,600	SAR 230,400	SAR 369,000
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Even under the conservative scenario, the identified value is dominated by routine time savings — meaning the business case does **not** depend on the harder-to-prove fraud number to be worthwhile. The fraud-escalation improvement is a genuine additional upside, not the load-bearing assumption.

7. What This Analysis Deliberately Does Not Claim

- It does not claim the AI agent is more *accurate* than a trained human triager — only faster and more consistent on volume.
 - It does not include the cost of running the system (LLM API costs, engineering time, monitoring) — a complete business case would net these out. At current LLM API pricing and ~12,000 tickets/month, this cost is small relative to the time-savings figure above, but should be measured, not assumed, before sign-off.
 - It does not address false positives/negatives in classification — the evaluation harness (companion project) is where that accuracy question is actually answered, and its results should feed back into this model before treating these numbers as final.
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8. Recommended Next Steps (if this were a real decision)

1. Replace the assumptions in Section 2 with actual ticket volume and agent-cost data.
 2. Run the evaluation harness against a sample of real historical tickets to get a measured accuracy/escalation-precision number, not an assumed one.
 3. Pilot on the 55% “routine” segment only for the first phase — the highest-confidence, lowest-risk value driver — before extending scope.
 4. Track two KPIs post-launch: average time-to-escalation for fraud-flagged tickets, and agent hours spent on triage vs. resolution.
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This document is a business analysis built on top of a working technical prototype — the code, tests, and design decisions behind the agent itself are documented separately in the [project README](#).

